

JOHN DEREK ANAS

Senior Research Engineer | AI-Driven Smart Buildings & Full-Stack Systems

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PROFESSIONAL SUMMARY

Senior Research Engineer with 10+ years across AI/ML research, full-stack engineering, and applied product delivery. Currently leads technical R&D on adaptive AI for sustainable buildings: non-intrusive, hardware-agnostic HVAC control that deploys into existing Fan Coil Unit (FCU) networks to cut energy use and carbon emissions at campus and commercial scale. Proven record shipping AI platforms (LLM fine-tuning, RAG, computer vision) and scalable TypeScript/React applications. AWS and GCP certified, hands-on across the full delivery lifecycle.

TECHNICAL SKILLS

AI & Machine Learning: LLM fine-tuning (GGUF), Retrieval-Augmented Generation (RAG), computer vision, OpenAI API and Agents SDK, AWS Bedrock, Google Vertex AI, PyTorch, Model Context Protocol (MCP), multi-agent systems

Smart Building & IoT: HVAC control systems, Fan Coil Unit (FCU) networks, Building Management Systems (BMS), BACnet, Modbus, MQTT, edge-to-cloud telemetry, energy and carbon optimization

Languages & Frameworks: TypeScript, JavaScript, Python, PHP, React.js, Node.js, Django, REST APIs

Cloud & DevOps: AWS, Google Cloud Platform (GCP), Docker, Terraform, CI/CD, containerized deployment

Data & Product: PostgreSQL, time-series pipelines, design systems, UX/UI (Figma), server-side rendering

PROFESSIONAL EXPERIENCE

Senior Research Engineer I

Jun 2026 – Present

Nanyang Technological University (NTU), Smart Building Initiatives | Singapore

- Lead technical R&D for a scalable, non-intrusive HVAC control framework that integrates into existing Fan Coil Unit (FCU) networks with minimal retrofit.
- Architect a hardware-agnostic, plug-and-play deployment model for rapid rollout across heterogeneous BMS and BACnet environments.
- Drive AI-led energy optimization that reduces carbon emissions and energy overhead across campus and commercial building grids.
- Lead a cross-functional engineering team turning applied research into production-grade control intelligence.

Research Engineer II

2024 – 2026

Nanyang Technological University (NTU-CCDS) | Singapore

- Shipped MazeMate, a 3D gamified Data Structures and Algorithms learning platform in React.js and TypeScript, with a fine-tuned OpenAI assistant and RAG knowledge base.
- Developed AI/ML models for real-time risk forecasting, crew well-being monitoring, and maritime safety decision support.
- Built AI-driven interactive research tools and championed open-source LLM adoption within research workflows.

Software Engineer

2019 – 2024

SPH Media | Singapore

- Revamped multiple magazine websites in React.js and TypeScript, improving user engagement.
- Implemented server-side rendering via the CUE content system to speed up content delivery.
- Migrated monolithic CMS properties into a shared TypeScript content-as-a-service monorepo.

Campaign Manager / Developer

2017 – 2019

Teckwah Industrial | Singapore

- Built cross-media campaigns with XMPie variable data publishing, plus e-commerce and Web-to-Print platforms in PHP, Python, and AngularJS.

Full Stack Developer

2014 – 2017

Flava Design | Singapore

- Delivered full-stack WordPress and Drupal projects with custom CMS and CRM features on CodeIgniter, plus mobile apps and UI/UX design work.

EDUCATION

Bachelor of Science in Information Technology

2010 – 2014

De La Salle University, Philippines

CERTIFICATIONS

Cloud & AI: Google Cloud Gen AI Academy APAC (Google, 2026); AWS Certified Cloud Practitioner (2022)

Machine Learning: PyTorch Essential Training (LinkedIn, 2025); MCP and AI Agents Fundamentals (Hugging Face, 2025)

Practice & Research: ITIL Foundation v4 (PeopleCert, 2023); Social and Behavioral Research (CITI Program, 2025)